POLITECNICO MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE E DELL'INFORMAZIONE

IoT Challenge 3
PART 2 - Exercise LoRaWAN

Tesi di Laurea Magistrale in Telecommunication Engineering

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1.1. EQ1

Network Parameters:

- Carrier Frequency: 868 MHz(EU868)
- Bandwidth: 125 kHz
- Number of Nodes (N): 50
- Transmission Rate (λ): 1 $packet/minute$
- Packet Size: 39 bytes
- Transmission Power: 14 dBm
- Transmission Interval: $\lambda = 1 packet/minute$
- Success Rate Target: At least 70%

Success Rate Formula: According to the ALOHA protocol, the success rate $p_{success}$ is calculated as:

$$P_{\text{success}} = e^{-2G}$$

Where G is the load parameter, representing the system's load, defined as:

$$G = N \cdot \lambda \cdot t$$

From Table 1.1, it can be observed that as the Spreading Factor (SF) increases, the transmitted airtime (i.e., the duration of each packet's transmission) gradually increases, while the success rate steadily decreases. Although a higher SF can extend the signal's range, it also reduces the data transmission rate, which leads to more overlap in time windows with other nodes and thus lowers the success rate.
Specifically, **SF 8** (success rate: **73.5%**) is the highest SF configuration that still meets the 70% success rate requirement.

SF	AirTime(ms)	Success Rate
SF7	102.7	84.3%
SF8	184.8	**73.5%**
SF9	328.7	57.8%
SF10	616.4	35.8%
SF11	1314.8	11.2%
SF12	2465.8	1.6%

Table 1.1: EQ1:Success Rate

1.2. EQ2

As shown in Figure 1.1, the system collects ambient temperature and humidity data and transmits it wirelessly via LoRaWAN to the ThingSpeak platform for remote monitoring and visualization.

Since physical hardware is not available during the design, we simulate the system behavior using Node-RED.

Example Output:

```
"temperature": 25.3,
"humidity": 48.7
```

A system block diagram will be sketched in Node-RED to describe the complete data flow.

Arduino MKR WAN 1310 (Function Node) Function:

Simulates data processing on Arduino. It reads the raw DHT22 values, packs them into a JSON object, and prepares the data for LoRaWAN transmission. **LoRa Transmission (Function Node) Function:**

Simulates the LoRaWAN transmission process. The data packet generated by Arduino is encoded and "sent" through a virtual LoRaWAN channel to the Gateway.

Gateway Receiver + HTTP (Function Node) Function:

Simulates a LoRaWAN Gateway receiving the packet. It decodes the incoming LoRa message, extracts the payload, and formats it properly to be uploaded to ThingSpeak via HTTP API.

Debug Output (Debug Node) Function:

Displays the HTTP request response and intermediate data packets, facilitating troubleshooting and ensuring the whole process works correctly.

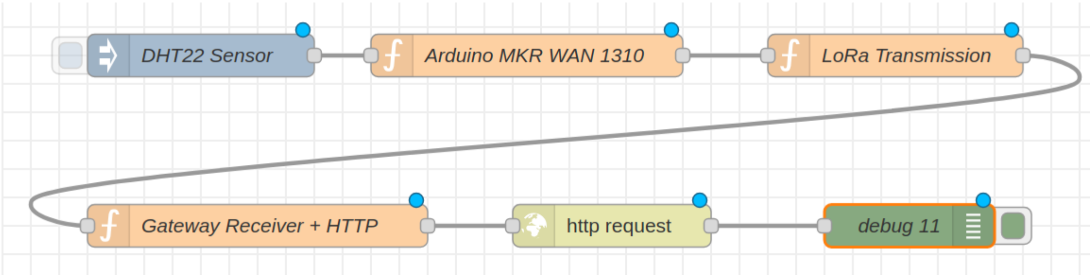

Figure 1.1: EQ2: System Block Diagram

1.3. EQ3

This project aims to use the LoRaSim simulator to reproduce Figure 5 and Figure 7 from the paper "Do LoRa Low-Power Wide-Area Networks Scale?", and to deeply understand the scalability issues of LoRa networks under different node densities and configurations through simulation experiments.
In the LoRaSim simulation, the Data Extraction Rate (DER) is calculated using the following formula:

$$\mathrm{DER} = \frac{\mathrm{nrTransmissions} - \mathrm{nrCollisions}}{\mathrm{nrTransmissions}}$$

where:

- nrTransmissions: total number of packet transmission attempts by all nodes,
- nrCollisions: number of transmission collisions detected.

The DER reflects the ratio of successfully received packets over the total number of transmission attempts. A higher DER indicates better network performance.

1.3.1. Figure 5 Reproduction

Figure 5 mainly explores the impact of different scheduling and optimization strategies (exp0-exp4) on the Data Extraction Rate (DER) under a single gateway deployment. The goal of the paper is to investigate whether it is possible to support more nodes without adding additional gateways, by using smart optimizations.

- Simulation duration: 1 day (i.e., $24 \times 3600 \times 1000$ milliseconds)
- Transmission rate: 1,000,000 bps
- Number of nodes: Increase from 1 to 1600
- simulate exp3, exp4, and exp5 optimization strategies

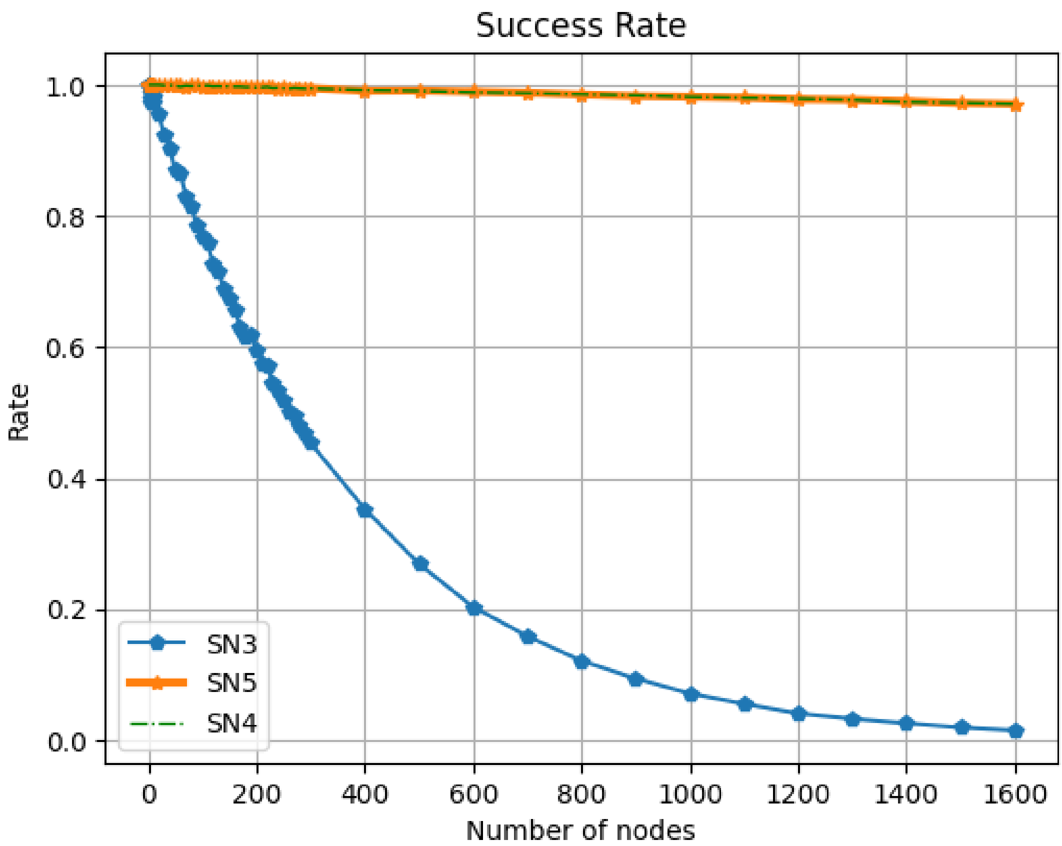

Figure 1.2: EQ3:Figure 5 Reproduction

SN3	Fixed parameters: All nodes use the same Spreading Factor (SF), Bandwidth (BW), and Transmission Power (TP). No adaptation based on distance to the gateway.
SN4	Dynamic SF adaptation: Nodes dynamically select Spreading Factor (SF) based on their distance to the gateway. Farther nodes use higher SFs, closer nodes use lower SFs. Other parameters like TP and BW might also adjust.
SN5	Full optimization: Dynamically adjust not only SF but also TP and BW to minimize transmission time and collisions, and maximize network capacity. It's a more aggressive optimization strategy.

Table 1.2: SN description

SN3 curve drops quickly: Due to large spreading factors (SF) and long airtime, collisions happen frequently, and the network becomes highly loaded. Static settings can only support about 50 nodes (DER > 0.9).
Dynamic minimum Airtime or dynamic Airtime+Power optimization can significantly improve network capacity. Because most nearby nodes use low spreading factors (high data rates), the airtime is greatly reduced, thus lowering the risk of collisions. With dynamic parameter optimization (either minimum airtime (SN4) or airtime plus power optimization (SN5)), the network can support over 1600 nodes while maintaining DER > 0.9.
At the current node density, the key factor influencing DER is the dynamic selection of SF, and since SN4 and SN5 use similar SF strategies, their DER curves almost overlap.
Figure 1.2 shows that by combining multiple optimization strategies, a LoRa network can significantly increase its supported number of nodes even under a single gateway.

1.3.2. Figure 7 Reproduction

This experiment analyzes how the Data Extraction Rate (DER) changes as the number of sinks increases under fixed communication settings (SN1) in a LoRa network. The number of sinks M is gradually increased (from 1 to 8 to 24, etc.).
Increasing the number of sinks significantly improves DER. For example, with 200 nodes,

```
for n_nodes in node_list:
    for base_stations in [1, 2, 3, 4, 8, 24]:  # Set sinks
        simulate(n_nodes, tx_rate, 0, duration, sinks, 1)
```

one sink cannot meet the typical DER>0.7 requirement, but three sinks can. With 24 sinks, even more than 400 nodes can maintain a DER above 0.9.
A node's packet only needs to be successfully received by one of the sinks.
The more sinks there are, the higher the chance of successful packet reception, thus

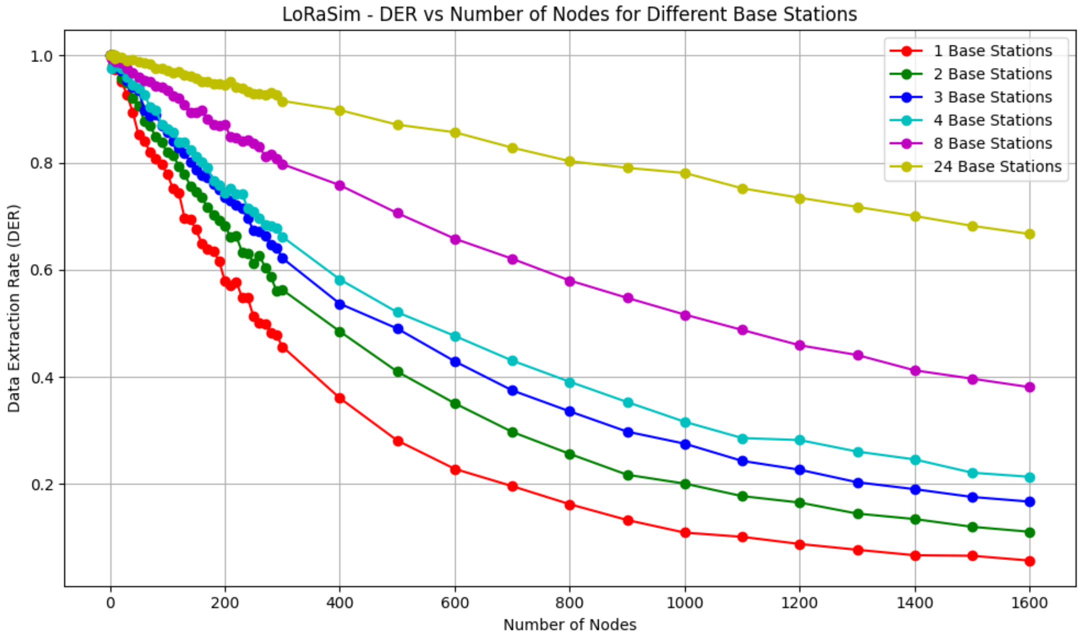

Figure 1.3: Figure 7 Reproduction (SN1)

effectively reducing packet loss. Multiple sinks provide spatial diversity, meaning that signals from nodes are captured at different physical locations, further enhancing network reliability.

This experiment demonstrates that under fixed communication settings (SN1), increasing the number of sinks can effectively and significantly enhance the DER in LoRa networks. Multiple sinks bring advantages such as enhanced capture effect, spatial diversity, and the sufficiency of a single successful reception, greatly improving the success rate of data transmission.